Experiment 09

Implementation of SUBNETTING in CISCO PACKET TRACER simulator

Aim

To design multiple subnets, assign static IP addresses, connect them through routers and compare connectivity with/without routing.

Requirements

Packet Tracer; two 2911 routers (three Gigabit interfaces each), four access switches and five hosts per LAN.

Procedure / algorithm

1. Calculate /27 from a /24: borrow three bits; obtain eight subnets, block size 32 and 30 usable hosts. Network/broadcast addresses are not host addresses.

2. Build four separate LANs 192.168.1.0/27 through 192.168.4.0/27, as in the manual's address examples. Each has a separate switch.

3. Assign host addresses .11-.15, /27 masks and gateway .1 from addressing.csv. R1 serves LANs 1/2 and R2 serves LANs 3/4.

4. Connect R1/R2 through the added 10.255.0.0/30 transit link. Apply both router configurations, including reciprocal remote-LAN static routes.

5. Apply access-port switch configuration to each LAN switch. Do not assign network IDs to physical Layer 2 switch ports.

6. Ping within a LAN, between the two LANs on one router and across the transit link. Inspect the routing table and packet simulation.

7. Disable the relevant router/transit link and compare failed inter-subnet communication with continuing same-subnet communication.

Complete source / configuration: addressing.csv

device	interface	ip	prefix	gateway
R1	GigabitEthernet0/0	192.168.1.1	27	none
LAN1-PC1	FastEthernet0	192.168.1.11	27	192.168.1.1
LAN1-PC2	FastEthernet0	192.168.1.12	27	192.168.1.1
LAN1-PC3	FastEthernet0	192.168.1.13	27	192.168.1.1
LAN1-PC4	FastEthernet0	192.168.1.14	27	192.168.1.1
LAN1-PC5	FastEthernet0	192.168.1.15	27	192.168.1.1
R1	GigabitEthernet0/1	192.168.2.1	27	none
LAN2-PC1	FastEthernet0	192.168.2.11	27	192.168.2.1
LAN2-PC2	FastEthernet0	192.168.2.12	27	192.168.2.1
LAN2-PC3	FastEthernet0	192.168.2.13	27	192.168.2.1
LAN2-PC4	FastEthernet0	192.168.2.14	27	192.168.2.1
LAN2-PC5	FastEthernet0	192.168.2.15	27	192.168.2.1
R1	GigabitEthernet0/2	10.255.0.1	30	none
R2	GigabitEthernet0/0	192.168.3.1	27	none
LAN3-PC1	FastEthernet0	192.168.3.11	27	192.168.3.1
LAN3-PC2	FastEthernet0	192.168.3.12	27	192.168.3.1
LAN3-PC3	FastEthernet0	192.168.3.13	27	192.168.3.1
LAN3-PC4	FastEthernet0	192.168.3.14	27	192.168.3.1
LAN3-PC5	FastEthernet0	192.168.3.15	27	192.168.3.1
R2	GigabitEthernet0/1	192.168.4.1	27	none
LAN4-PC1	FastEthernet0	192.168.4.11	27	192.168.4.1
LAN4-PC2	FastEthernet0	192.168.4.12	27	192.168.4.1
LAN4-PC3	FastEthernet0	192.168.4.13	27	192.168.4.1
LAN4-PC4	FastEthernet0	192.168.4.14	27	192.168.4.1
LAN4-PC5	FastEthernet0	192.168.4.15	27	192.168.4.1
R2	GigabitEthernet0/2	10.255.0.2	30	none

Complete source / configuration: R1.cfg

```
enable
configure terminal
hostname R1
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.224
 no shutdown
 exit
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.224
 no shutdown
 exit
interface GigabitEthernet0/2
 ip address 10.255.0.1 255.255.255.252
 no shutdown
 exit
ip route 192.168.3.0 255.255.255.224 10.255.0.2
ip route 192.168.4.0 255.255.255.224 10.255.0.2
end
copy running-config startup-config
```

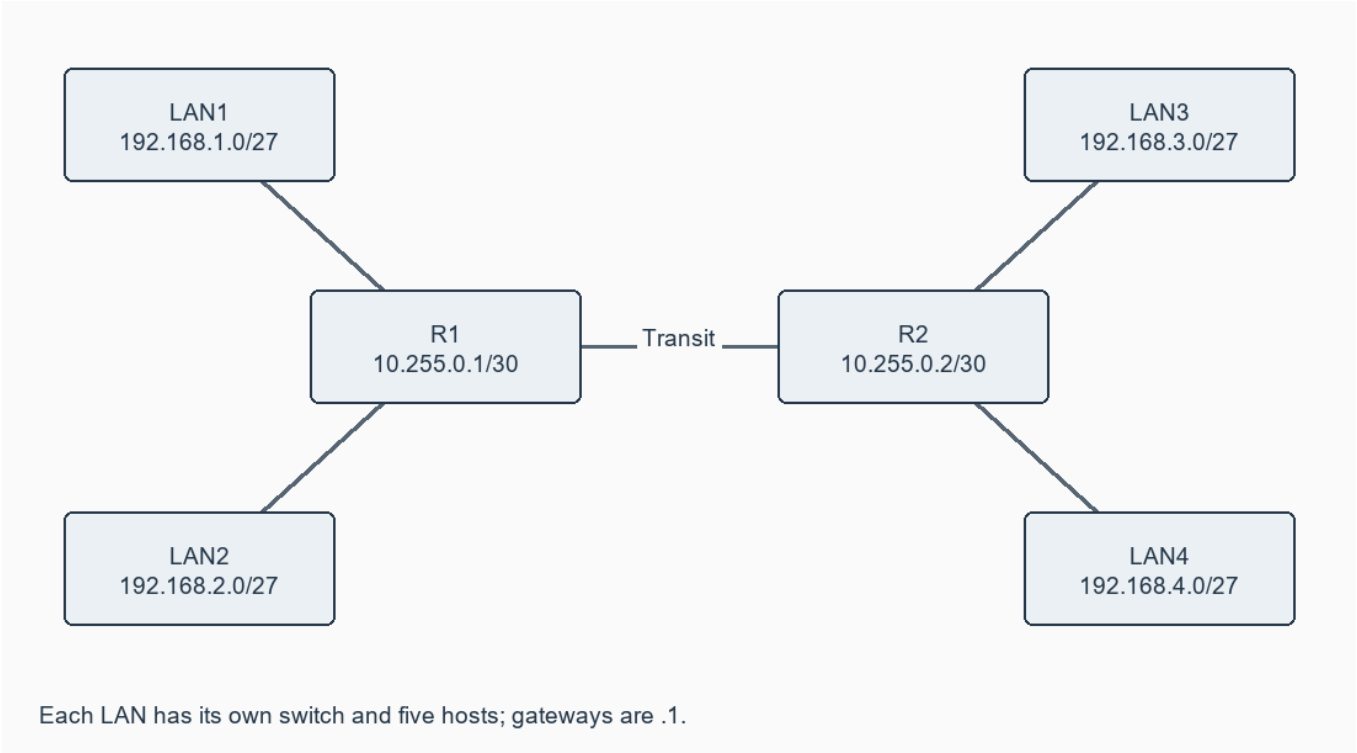
Complete source / configuration: R2.cfg

```
enable
configure terminal
hostname R2
interface GigabitEthernet0/0
 ip address 192.168.3.1 255.255.255.224
 no shutdown
 exit
interface GigabitEthernet0/1
 ip address 192.168.4.1 255.255.255.224
 no shutdown
 exit
interface GigabitEthernet0/2
 ip address 10.255.0.2 255.255.255.252
 no shutdown
 exit
ip route 192.168.1.0 255.255.255.224 10.255.0.1
ip route 192.168.2.0 255.255.255.224 10.255.0.1
end
copy running-config startup-config
```

Complete source / configuration: switch.cfg

```
enable
configure terminal
interface range FastEthernet0/1-6
 switchport mode access
 no shutdown
end
copy running-config startup-config
```

Diagram: subnets



Each LAN has its own switch and five hosts; gateways are .1.

Actual output / verification

Not executed in the required lab/tool environment. No actual operational output is claimed.

Expected output - not an execution record

Same-subnet traffic works through the switch. Inter-subnet traffic requires reachable gateways/routes. Removing routing breaks inter-subnet traffic but not local traffic.

Observations

The manual mixes a /24-to-/27 calculation with LANs from four different /24 blocks; this submission explicitly distinguishes them. Separate switches avoid unintended shared broadcast domains. The transit link/routes complete cross-router connectivity that the manual omits.

Result

A complete four-subnet configuration is supplied and address plans are checked offline; Packet Tracer connectivity remains pending.